
EE 305: Digital Signal Processing

Course Code	EE 305
Course Name	Digital Signal Processing
L-T-P-C	3-1-0-4
Prerequisites	IC-260 - Signals and Systems
Distribution	Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **Discrete time signals and systems:** Types of systems, LTI systems and their properties, impulse response and convolution, Difference equations, Eigen-functions of LTI systems [4 Lectures]
- **Discrete time signal transform:** Discrete time Fourier Transform (DTFT) and examples, Properties, Convergence of signals, Z-transform and examples, Properties, Difference equation representation, Inverse Z-transform. [4 Lectures]
- **Sampling:** Time domain and frequency domain representation, Nyquist theorem, Signal reconstruction, Discrete-time processing of continuous-time signals, Continuous-time processing of discrete-time signals, Changing the sampling rate, Multi-rate signal processing, Sub-Nyquist sampling and its applications. [10 Lectures]
- **Filtering and Frequency response of LTI systems:** Discrete-time frequency selective filtering, Phase distortion and delay, Characterization with difference equations, Stability and Causality, Frequency response of rational system functions, All pass and minimum-phase systems, Basics of filter design, Z-transform characterization of IIR filters, Window functions for FIR filters, Filter structures for IIR and FIR filters. [10 Lectures]
- **Discrete Fourier transform (DFT):** Discrete Fourier series and its properties, Fourier transform of periodic signals, Sampling the Fourier transform, DFT and its properties, Linear and circular convolution, Efficient computation of DFT using the Fast Fourier transform (FFT). [10 Lectures]
- **Fourier analysis of signals using the DFT:** Pipeline for analyzing continuous time signals, Effect of windowing, Effect of spectral sampling. [4 Lectures]

Text Books

1. Text for Unit 1 to Unit 6: Alan V. Oppenheim, Ronald W. Schaffer, John R. Buck., **Discrete-Time Signal Processing**, 2nd Edition, Pearson, 1999.

Additional reference

1. John G. Proakis, Dimitris G. Manolakis., **Digital Signal Processing Principles, Algorithms, and Applications**, 4th Edition, Pearson 2007.